Biology Class 11 National Book Foundation FBISE Quality Notes
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Chapter 5: Acellular Life

5.1 Viruses - status, history and structure (SLO 1, 2)

- Viruses are pathogens of animals and plants. They are **not cells**, cannot replicate independently, cannot make their own energy and proteins, and are too small for the light microscope. Study of viruses = **virology**.

Living characters	Non-living characters
Occur in different varieties or strains	Lack cellular structure, coenzymes and enzyme system; no metabolism of their own
Have genetic material (DNA or RNA) which can mutate	Can be crystallized and stored in bottles
Reproduce using metabolic machinery of host	Do not respire
Intracellular obligate parasites causing disease	Behave as inert infectious particles outside the host
Destroyed by ultraviolet rays	

- So viruses show characters of both living and non-living things.

History and discovery

- "Virus" = Latin **venom** (poison). **Charles Chamberland** (1884, with Pasteur): rabies agent passed through a porcelain filter (pore 100-1000 nm) that stops bacteria.
- **Iwanowsky** (1892): filtered juice of tobacco mosaic diseased leaves still infected healthy plants. By 1900 such "**filterable viruses**" were known in plants and animals.
- **W. M. Stanley** (1935) crystallized the infectious particle, tobacco mosaic virus (TMV). Later viruses were seen with the electron microscope.

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General structure of a virus (SLO 4)

- Complete viral particle = **virion**. Two parts: **core** and **coat**.
- **Core:** genome (DNA or RNA, single or double stranded) + enzymes. All ssRNA viruses have an enzyme to make double stranded RNA; **retroviruses and hepatitis B** have **reverse transcriptase** (RNA to DNA).
- **Coat:** capsid and (sometimes) envelope. **Capsid** = protein coat made of identical **capsomeres** (number is specific: herpes 162, adenovirus 252). Symmetry: **icosahedral** (20 triangles, polyhedral) or **helical** (hollow coil, rod shaped).
- **Envelope** (few viruses): lipoprotein, from host cell membrane + viral proteins; glycoprotein spikes help recognize the host.

Classification of viruses (SLO 3)

- **Baltimore classification** (David Baltimore, Nobel laureate): seven groups based on mode of replication and genome type.

By host	Examples
Bacteriophage	Attacks bacteria; DNA virus with polyhedral head and tail.
Plant viruses	Over 2000 known; mostly RNA genome (TMV, rod shaped; potato yellow dwarf virus).
Animal viruses	Foot and mouth disease, Rous sarcoma (cancer); papovirus (warts), poxvirus (smallpox), picornavirus (polio, hepatitis A), paramyxovirus (measles, mumps).

By structure	Examples
Helical capsid	Tobacco mosaic virus
Polyhedral capsid	Adenoviruses
Enveloped	Influenza viruses
Complex capsid	Bacteriophage

By genome	Examples
dsDNA	Smallpox virus
ssDNA	Mild rash virus

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By genome	Examples
dsRNA	Diarrhoea virus
ssRNA (acts as mRNA)	Rubella virus
ssRNA (template for mRNA)	Influenza virus
ssRNA (template for DNA)	HIV

5.2 Parasitic nature of viruses (SLO 5, 6, 7)

- Viruses are highly **host specific**: phage infects only bacteria, TMV only tobacco, rabies only mammals; HIV enters certain white blood cells, poliovirus reproduces in spinal nerve cells, hepatitis viruses in liver cells, cold viruses in upper respiratory lining.
- **Host range** is determined by specificity of attachment: protein spikes on the virus bind specific receptors on the host. Polio: humans and other primates only; rabies: all mammalian cells.
- **Why a host cell is a must (SLO 5):** viruses are **obligate intracellular parasites**; they lack metabolic enzymes and ribosomes for making proteins, so they cannot multiply outside a living cell.
- **Survival inside the host (SLO 6):** virus hides from the host immune system, rapidly changes its genetic constitution so vaccines / antibodies become ineffective, and the **capsid protects the genome from host nucleases** (enzymes that break DNA / RNA).
- **Unfavourable conditions outside host (SLO 7):** they need no food and remain **dormant** (no metabolism); their genome remains viable for long. Only pH and temperature (which denature protein) threaten them. Non-enveloped viruses survive longer; enveloped viruses survive shorter because the envelope is sensitive to sunlight and cleaning.

5.3 Bacteriophage (SLO 4, 8, 9)

Bacteriophage (phage): The virus that attacks bacteria. Tadpole shaped: head, neck and tail.

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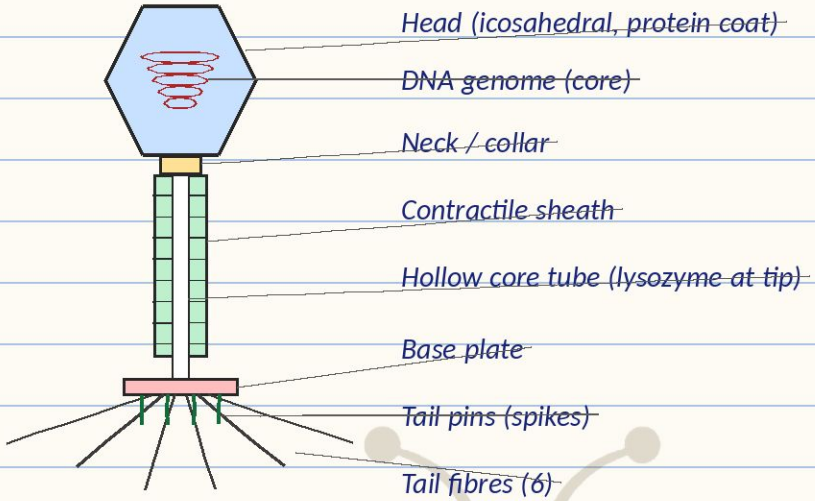


Fig: Structure of bacteriophage (tadpole shaped)

- ***Head:*** *icosahedral, core of (single stranded) DNA genome.* ***Neck / collar*** *separates head and tail.* ***Tail:*** *hollow tube through which nucleic acid passes, surrounded by* ***contractile sheath****.* ***Base plate*** *with about 6* ***tail fibres*** *and several* ***tail pins****, which bind to the bacterium.* ***Lysozyme*** *at the bottom of core tube digests part of host cell wall.*

Infection process (common to both cycles)

1. ***Adsorption:*** *tail fibres and pins attach to specific receptors on the bacterial cell.*
2. ***Penetration:*** *sheath contracts, lysozyme digests part of the envelope, hollow core tube is pushed through.*
3. ***Genome injection:*** *viral DNA enters the cytoplasm; the rest of the phage stays outside.*

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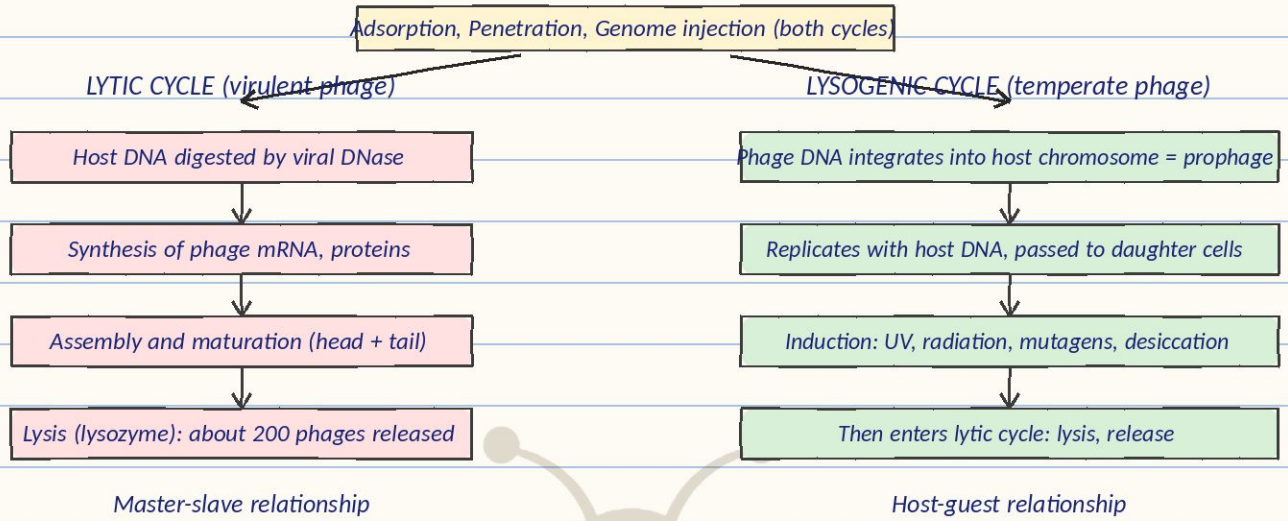


Fig: Lytic and lysogenic life cycles of bacteriophage

Lytic cycle (virulent phage)	**Lysogenic cycle (temperate phage)**
Immediately lyses host cell after multiplication	Phage DNA integrates in host chromosome as prophage (lysogeny); bacteria = lysogenic bacteria
Master-slave relationship: host DNA disintegrated by viral DNase; viral DNA takes control	Host-guest relationship: prophage replicates with host chromosome and passes to daughter cells
Steps: synthesis of phage components, assembly, maturation, lysis and release (20-25 min, about 200 phages)	Can multiply by lytic cycle or stay dormant; terminated by **induction** (desiccation, UV / ionizing radiation, mutagenic chemicals), then lytic cycle starts
Lysozyme accumulates, bacterium bursts	Prophage genes can give new properties (phage conversion), e.g. toxin genes of Clostridium botulinum

Usage of bacteriophage in genetic engineering (SLO 9)

- ***Vector:*** *besides plasmids, phage DNA is used as vector, e.g. to build a* ***genomic library*** *(collection of bacteria / phage clones with copies of all genes of a genome).*
- ***Phage therapy:*** *genetically engineered phages kill pathogenic bacteria.*

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Advantage over antibiotics: narrow spectrum, so harmful bacteria are killed without harming normal (good) flora.

- ***Nano medicine:*** *phages formulated as targeted drug-delivery vehicles.*

5.4 Human Immunodeficiency Virus (SLO 4, 10-16)

- *HIV causes **AIDS** (acquired immune deficiency syndrome). Identified in 1984 by Pasteur Institute (France) and NIH (USA); named HIV in 1986.*

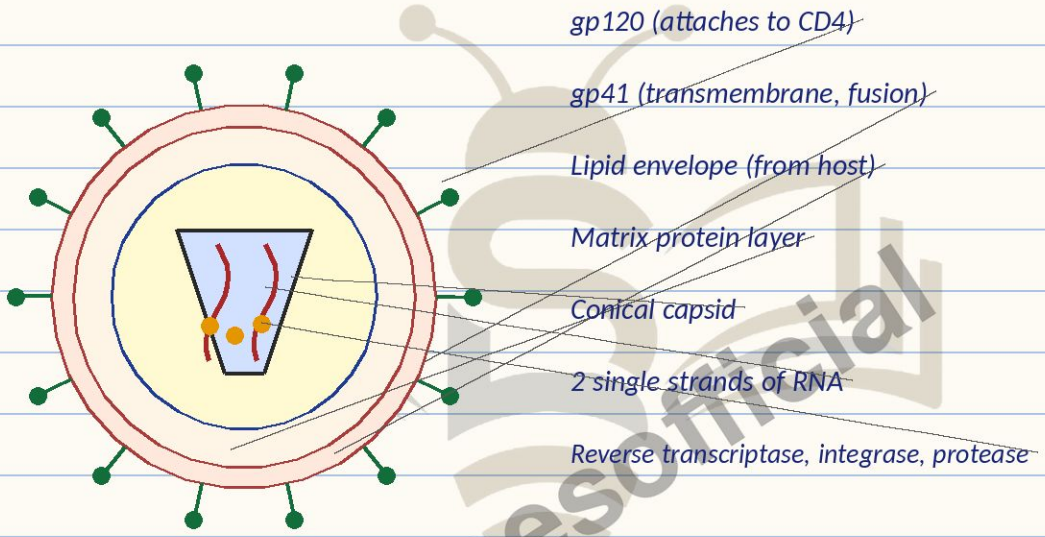


Fig: Structure of HIV (retrovirus)

- ***Structure:*** *HIV is a **retrovirus**, spherical. Outer **lipoprotein envelope** (double lipid layer from host during budding) with spikes: outer glycoprotein **gp120** (attaches to host cell) and transmembrane **gp41** (cell fusion). Beneath is a protein shell of **matrix proteins**; then a somewhat conical **capsid** of capsomeres. Core: **two single strands of RNA** + enzymes **reverse transcriptase** (RNA to DNA), **integrase** (joins viral DNA into host DNA) and **protease** (cuts large proteins).*

Specificity of HIV (SLO 12)

- *HIV attacks only **CD4 cells** (helper T cells, macrophages, certain brain cells) because gp120 recognizes and binds the **CD4 receptor** only. So it has host cell specificity.*

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Life cycle of HIV (SLO 10)

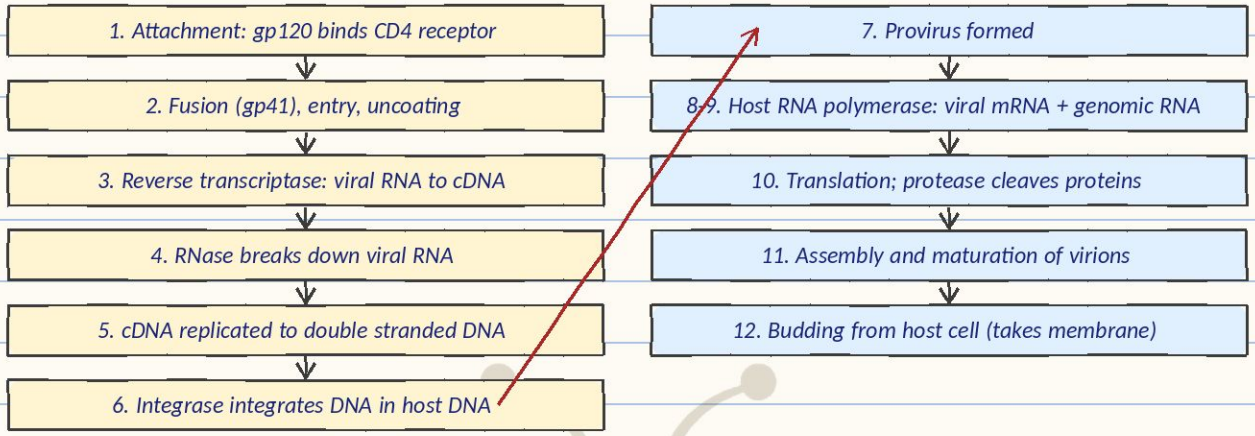


Fig: Life cycle of HIV

- *(1) gp120 binds CD4. (2) Envelope fuses using gp41; virion enters by endocytosis; **uncoating** leaves ssRNA and enzymes. (3) Reverse transcriptase makes single stranded **cDNA**. (4) RNase breaks down viral RNA. (5) cDNA replicated into double stranded cDNA. (6) Integrase integrates it into host DNA; (7) integrated DNA = **provirus**. (8-9) Host RNA polymerase makes viral mRNA and genomic RNA. (10) Host ribosomes translate mRNA; viral protease cleaves proteins to structural proteins. (11) Assembly and maturation. (12) Virions leave by **budding**, taking host membrane; host cell shrinks and becomes non-functional.*

Why "Human Immunodeficiency Virus" (SLO 11)

- ***Helper T cells** are the most important cells of immunity: they activate B cells to secrete antibodies and activate cytotoxic T cells to kill virus infected and cancer cells; they regulate immunity by enhancing other immune cells. HIV decreases helper T cells, causing deficiency of the human immune system, hence the name.*

Symptoms of AIDS (SLO 13)

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Stage	Features
Asymptomatic carrier	Fever, chills, aches, swollen lymph glands, itchy rash; disappear; no symptoms for 9 months or longer. Highly infectious; antibody blood test becomes positive.
AIDS related complex (ARC)	Swollen lymph glands (neck, armpit, groin) for months, night sweats, persistent cough, flu, persistent diarrhoea, memory loss, depression.
Full blown AIDS	Final stage: severe weight loss, weakness due to diarrhoea, usually one or more opportunistic infections.

Opportunistic diseases (SLO 14)

- HIV itself kills nobody; it only destroys T cells. With a weak defence system the patient is attacked by **opportunistic diseases**: skin cancer, fungal and viral infections, gastrointestinal, respiratory, nervous system and eye diseases.

Treatment (SLO 15)

- **Antiretroviral therapy (ART):** combination of medicines. Not a cure but controls the virus, lets a patient live longer and healthier, and reduces transmission. If started early, lifespan is close to HIV-negative people. Limitations: toxicities, costly, strict daily pill taking; so a cure is still the research goal.

Control measures (SLO 16)

1. Do not use used syringes and needles.
2. Blood for transfusion must be screened for HIV.
3. Do not share toothbrushes, blades, towels; take care at barber shops and salons.
4. Surgical instruments must be properly sterilized.
5. Avoid illicit sexual activity (AIDS is primarily sexually transmitted); follow Islamic teachings.
6. HIV positive mothers should not feed their babies.

- HIV is **not** spread by handshakes, hugging, coughing, sneezing, swimming pools, toilets, door knobs, telephones, furniture, or mosquitoes and insects.

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5.5 Viral diseases (SLO 17, 18)

Disease	Cause / transmission / symptoms / treatment / prevention
Hepatitis A (infectious)	HAV. Faecal-oral route. Fever, loss of appetite, nausea, vomiting, jaundice, dark urine, pale faeces. No antiviral therapy. Vaccine (inactivated HAV) + hygiene.
Hepatitis B (serum)	HBV. Blood, sexual contact, mother to newborn. Symptoms like A. Alpha interferon. Effective vaccine; screen blood.
Hepatitis C	Hepatitis C virus. **Only via blood.** Symptoms like hepatitis B. Treatment: alpha interferon + ribavirin (chronic). **No vaccine**; screen blood transfusions.
Hepatitis D	Book: caused by a viroid (only human disease from a viroid); enters liver cell inside a hepatitis B capsid; blood / serum transfusion; more severe than B; treatment and prevention same as HBV.
Hepatitis E	HEV. Faecal-oral. No antiviral treatment, no vaccine; hygiene.

- **Hepatitis** = inflammation of the liver (acute or chronic), mostly viral.

Herpes simplex	Details
Oral herpes	HSV type 1. Saliva, contact with sores, shared toothbrushes / utensils. Blisters on lips, around mouth, face, tongue. Antiviral drugs. Avoid contact with affected area.
Genital herpes	HSV type 2. Sexual contact with someone infected. Sores on penis, vagina, buttocks, anus. Antiviral drugs. Avoid contact.

- Herpes = painful fluid-filled blisters on skin / epithelium; tingling, itching or burning first; sores last 7 to 10 days.

Disease	Details
Poliomyelitis	Polio virus (an enterovirus). Faecal-oral route. Replicates in oropharynx and intestine, spreads to blood and CNS, destroys motor neurons of spinal cord: paralysis (permanent). No antiviral; physiotherapy. Prevention: Salk (killed, IPV, injectable) and Sabin (live attenuated, OPV, oral) vaccines.

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Disease	Details
Cotton leaf curl disease (CLCuD)	Begomoviruses. Transmitted by **whitefly**. Deep downward cupping of youngest leaves, then cup-shaped leaf-like structures. Control: insecticides against the insect vector.

5.6 Economic losses due to viral infections (SLO 21)

- COVID-19 = greatest global health disaster of the century. Lockdowns slowed economic activity; companies closed or down-sized; job losses. Affected: services, manufacturing, agriculture, food, education, tourism, airlines, sports, entertainment; capital and labour markets, foreign trade, consumer spending, production, GDP.
- The pandemic cost the world more than **8.5 trillion dollars**. World Bank: even a weak influenza pandemic like H1N1 (2009) could reduce global GDP by about **300 billion dollars**.

5.7 Limitations of flu / cold vaccine (SLO 22)

- More than **200 viruses** cause the common cold, so a vaccine is not practical.
- Flu virus strains constantly evolve and the body's immune response changes over time, so the previous year's vaccine is useless against new strains.

5.8 Prions and viroids (SLO 19, 20)

- Acellular infectious particles simpler and smaller than viruses. Work of Nobel laureate **Stanley Prusiner** convinced biologists such agents exist.

Prions	Viroids
Infectious **protein** particles; no nucleic acid	Short, circular, single stranded RNA; no protein coat or envelope
Single protein encoded by a single cellular gene	RNA does not code for any protein; replicates using host RNA polymerase II
More resistant to UV and heat than viruses	Some are ribozymes (self-cleavage)

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Prions	Viroids
Clump in brain cells: memory loss, neurodegeneration, permanent damage	Spread by mechanical damage, seed, pollen, biological vectors
Diseases: Kuru (humans), mad cow disease (cattle); no treatment or cure	Diseases: potato spindle tuber, cucumber pale fruit; in humans (book) hepatitis D

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Chapter 5: Short Questions with Answers (SLO based)

Q1. [SLO 1] What are viruses? Why are they called acellular?

Ans: Viruses are infectious pathogens of animals and plants that are not cells; they have no cellular structure and cannot replicate independently.

Q2. [SLO 1] Why are viruses considered both living and non-living?

Ans: They have genetic material, mutate, show strains and reproduce inside a host (living); but they lack cell structure and enzymes, can be crystallized and do not respire (non-living).

Q3. [SLO 1] List the living characteristics of viruses.

Ans: Occur in strains, have DNA or RNA that can mutate, reproduce using host machinery, are intracellular obligate parasites, are destroyed by ultraviolet rays.

Q4. [SLO 1] List the non-living characteristics of viruses.

Ans: No cellular structure, no enzyme / coenzyme system or own metabolism, can be crystallized and stored, do not respire; inert outside the host.

Q5. [SLO 2] How did Chamberland contribute to the discovery of viruses?

Ans: In 1884 he found that the rabies agent passed through a porcelain filter that removes all bacteria, giving the first evidence of viruses.

Q6. [SLO 2] What did Iwanowsky observe?

Ans: In 1892 filtered juice of tobacco mosaic diseased leaves, free of bacteria, still caused disease in healthy leaves.

Q7. [SLO 2] What did W. M. Stanley do?

Ans: In 1935 he crystallized the infectious particle, the tobacco mosaic virus.

Q8. [SLO 2] What does the word virus mean?

Ans: It is Latin for venom, meaning poison.

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Q9. [SLO 4] What is a virion?

Ans: A complete viral particle.

Q10. [SLO 4] Differentiate between core and coat of a virus.

Ans: Core: viral genome + enzymes. Coat: capsid (protein) and sometimes envelope.

Q11. [SLO 4] What are capsid and capsomeres?

Ans: Capsid is the protein coat surrounding the core, made of identical repeating subunits called capsomeres (herpes 162, adenovirus 252).

Q12. [SLO 4] Name the two forms of symmetry of viral capsid.

Ans: Icosahedral (polyhedral: 20 triangles) and helical (hollow coil, rod shaped).

Q13. [SLO 4] What is the viral envelope?

Ans: A lipoprotein layer from host cell membrane with virus encoded glycoprotein spikes that help recognize the host.

Q14. [SLO 3] Classify viruses on the basis of their hosts.

Ans: Bacteriophages (attack bacteria), plant viruses (e.g. TMV) and animal viruses (e.g. polio, measles).

Q15. [SLO 3] Classify viruses on the basis of capsid.

Ans: Helical (TMV), polyhedral (adenovirus), enveloped (influenza), complex (bacteriophage).

Q16. [SLO 3] What is the Baltimore classification?

Ans: System by David Baltimore placing viruses into seven groups by mode of replication and genome type.

Q17. [SLO 5] Why must a virus have a host cell to complete its life cycle?

Ans: It is an obligate intracellular parasite that lacks metabolic enzymes and ribosomes for protein synthesis, so it cannot multiply outside a living cell.

Q18. [SLO 5] What determines the host range of a virus?

Ans: Specificity of attachment: virus protein spikes bind specific receptors on host cells.

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Q19. [SLO 6] How does a virus protect itself from the immune system inside a host?

Ans: It hides inside the cell, changes its genetic constitution rapidly so antibodies and vaccines fail, and the capsid protects the genome from nucleases.

Q20. [SLO 7] How does a virus survive outside the host?

Ans: It stays dormant (no metabolism, no food needed); only pH and temperature can denature its proteins; the genome stays viable for long.

Q21. [SLO 7] Why do enveloped viruses survive shorter outside the host?

Ans: The envelope is sensitive to degradation by sunlight and cleaning procedures.

Q22. [SLO 4] Describe the structure of a bacteriophage.

Ans: Tadpole shaped: icosahedral head with DNA, neck, hollow tail with contractile sheath, base plate with tail fibres and pins; lysozyme at tip of core tube.

Q23. [SLO 4] What is the role of tail fibres and lysozyme?

Ans: Tail fibres and pins bind the phage to the bacterium; lysozyme digests part of the host cell wall so the core tube can penetrate.

Q24. [SLO 8] Name the steps of the infection process of bacteriophage.

Ans: Adsorption, penetration and genome injection.

Q25. [SLO 8] What is a virulent phage?

Ans: A phage with lytic cycle that lyses the host immediately after multiplication.

Q26. [SLO 8] Why is the lytic cycle called a master-slave relationship?

Ans: Viral DNase disintegrates host DNA and viral DNA takes control of the host metabolic machinery.

Q27. [SLO 8] What are prophage and lysogeny?

Ans: Prophage is phage DNA integrated into the host chromosome; the process is lysogeny and the bacteria are lysogenic bacteria.

Q28. [SLO 8] What is induction?

Ans: Termination of the lysogenic state by adverse conditions (desiccation, UV or ionizing radiation, mutagenic chemicals), starting the lytic cycle.

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Q29. [SLO 8] Why is the lysogenic cycle called host-guest relationship?

Ans: Phage DNA is replicated with host chromosome and passed on to daughter cells without killing the host.

Q30. [SLO 8] How many phages are produced in a lytic cycle?

Ans: About 200 phage particles in 20 to 25 minutes.

Q31. [SLO 9] What is a genomic library?

Ans: A collection of bacteria or phage clones containing multiple copies of all genes of an individual's genome, made using phage DNA as a vector.

Q32. [SLO 9] What is phage therapy? Write its advantage.

Ans: Use of genetically engineered phages to kill pathogenic bacteria; being narrow spectrum, it does not harm normal flora like antibiotics do.

Q33. [SLO 4] What is HIV? Who identified it?

Ans: Human immunodeficiency virus, a retrovirus causing AIDS; identified in 1984 by Pasteur Institute (France) and NIH (USA), named HIV in 1986.

Q34. [SLO 4] What is the role of gp120 and gp41 of HIV?

Ans: gp120 attaches the virus to the host cell; gp41 is needed for fusion with the cell.

Q35. [SLO 4] Name the enzymes of HIV and their functions.

Ans: Reverse transcriptase (RNA to DNA), integrase (integrates viral DNA in host DNA), protease (cuts large proteins).

Q36. [SLO 12] Why does HIV attack only helper T cells?

Ans: gp120 recognizes only the CD4 receptor, present on helper T cells, macrophages and some brain cells.

Q37. [SLO 10] What is a provirus?

Ans: The viral DNA integrated into the host cell DNA.

Q38. [SLO 10] What is the role of reverse transcriptase?

Ans: It makes complementary DNA (cDNA) from viral RNA.

Q39. [SLO 10] How do new HIV virions leave the host cell?

Ans: By budding, enclosing a portion of host cell membrane; the host cell shrinks and becomes non-functional.

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Q40. [SLO 11] Why is the virus named Human Immunodeficiency Virus?

Ans: It reduces the number of helper T cells, which regulate immunity, causing deficiency of the human immune system.

Q41. [SLO 11] What is the role of helper T cells?

Ans: They activate B cells to secrete antibodies and cytotoxic T cells to kill infected and cancerous cells.

Q42. [SLO 13] List the three stages of HIV infection.

Ans: Asymptomatic carrier, AIDS related complex (ARC) and full blown AIDS.

Q43. [SLO 13] Write the symptoms of AIDS related complex.

Ans: Persistent swollen lymph glands, night sweats, persistent cough, flu, diarrhoea, memory loss and depression.

Q44. [SLO 14] What are opportunistic diseases?

Ans: Diseases attacking an AIDS patient because of the weak defence system, e.g. skin cancer, fungal, viral, respiratory, gastrointestinal, nervous and eye diseases.

Q45. [SLO 15] What is ART? Is it a cure?

Ans: Antiretroviral therapy, a combination of medicines that controls HIV; it is not a cure.

Q46. [SLO 16] List four precautions to prevent AIDS.

Ans: Do not use used syringes, screen blood for transfusion, do not share razors / toothbrushes / towels, sterilize surgical instruments, avoid illicit sex.

Q47. [SLO 16] How is HIV not transmitted?

Ans: By handshakes, hugging, coughing, sneezing, swimming pools, toilets, door knobs, telephones, mosquitoes or insects.

Q48. [SLO 17] Differentiate between hepatitis A and B.

Ans: A: HAV, faecal-oral, vaccine of inactivated virus, no antiviral. B: HBV, via blood, sexual contact and mother to newborn, alpha interferon, vaccine.

Q49. [SLO 17] Write the cause, transmission, treatment and prevention of hepatitis C.

Ans: Hepatitis C virus; transmitted only via blood; alpha interferon + ribavirin; no vaccine, screen blood transfusion.

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Q50. [SLO 17] What causes herpes? Name its types.

Ans: Herpes simplex virus: oral herpes (HSV type 1) and genital herpes (HSV type 2).

Q51. [SLO 17] Write symptoms and treatment of herpes.

Ans: Painful fluid-filled blisters (tingling, itching first) lasting 7-10 days; treated with antiviral drugs; avoid contact with affected area.

Q52. [SLO 17] What are the causes and vaccines of polio?

Ans: Polio virus (enterovirus), faecal-oral; vaccines: Salk (killed, injectable) and Sabin (live attenuated, oral).

Q53. [SLO 17] How does polio cause paralysis?

Ans: The virus replicates in motor neurons of the spinal cord; death of these cells causes permanent paralysis of the muscles they supply.

Q54. [SLO 17] What is cotton leaf curl disease? How is it controlled?

Ans: A disease of cotton by Begomoviruses spread by whitefly; deep downward cupping of leaves; controlled by insecticides against the vector.

Q55. [SLO 18] Name the transmission routes of hepatitis A, C, polio and herpes.

Ans: Hepatitis A and polio: faecal-oral; hepatitis C: blood; herpes: saliva / contact with sores (type 1), sexual contact (type 2).

Q56. [SLO 19] What are prions? Write their characteristics.

Ans: Infectious protein particles with no nucleic acid, smaller than viruses, more resistant to UV and heat, made of a single protein encoded by a cellular gene.

Q57. [SLO 19] What are viroids?

Ans: Pathogens of short circular single stranded RNA without protein coat or envelope; RNA does not code for protein.

Q58. [SLO 20] Name diseases caused by prions and viroids.

Ans: Prions: Kuru in humans, mad cow disease in cattle. Viroids: potato spindle tuber, cucumber pale fruit disease (and hepatitis D as stated in the book).

Q59. [SLO 21] How do viral infections cause global economic loss?

Ans: Lockdowns close businesses and cause job loss; services, agriculture, tourism, airlines, trade and GDP are hit; COVID-19 cost over 8.5 trillion dollars.

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Q60. [SLO 22] Why is a vaccine against the common cold / flu not practical?

Ans: More than 200 viruses cause cold, flu strains constantly evolve and the immune response changes, so last year's vaccine fails against new strains.

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Chapter 5: Long Questions with Answers (SLO based)

Q1. [SLO 1, 2] Justify the status of viruses among living and non-living things. Trace the history of their discovery.

Ans:

Living	***Non-living***
Strains; DNA / RNA that mutates	*No cellular structure, no enzymes*
Reproduce using host machinery	*Crystallized and stored in bottles*
Obligate intracellular parasites	*Do not respire; inert outside host*

- *Hence viruses show characters of both. History: Chamberland (1884) rabies agent passed porcelain filter; Iwanowsky (1892) tobacco mosaic filtered juice infected healthy leaves; filterable viruses (1900); Stanley (1935) crystallized TMV; seen later with electron microscope.*

Q2. [SLO 3, 4] Classify viruses on the basis of host and structure. Describe the general structure of virus.

Ans:

- *Host: bacteriophage, plant viruses (TMV), animal viruses (polio, measles, smallpox).*
- *Capsid: helical (TMV), polyhedral (adenovirus), enveloped (influenza), complex (bacteriophage). Genome: dsDNA, ssDNA, dsRNA, ssRNA (three kinds, e.g. rubella, influenza, HIV).*
- *Structure: virion = core (DNA or RNA genome + enzymes like reverse transcriptase) + coat (protein capsid of capsomeres with icosahedral or helical symmetry; some have lipoprotein envelope with glycoprotein spikes).*

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Q3. [SLO 4] Describe the structure of a model bacteriophage with a labelled diagram.

Ans:

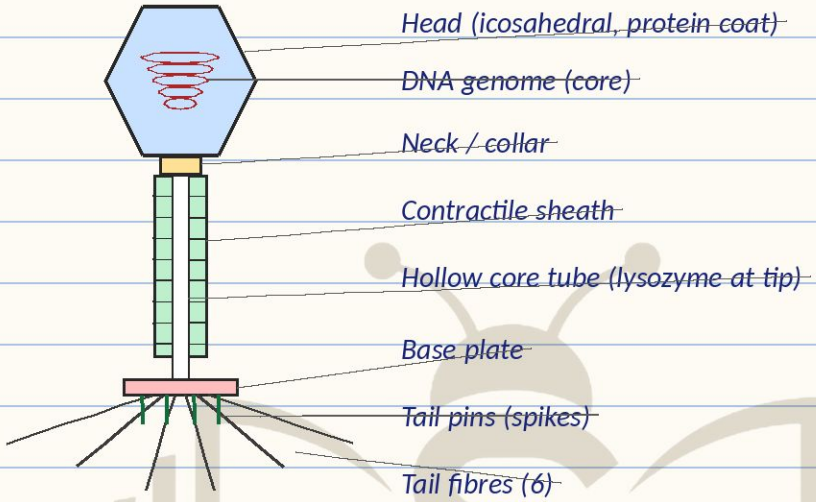


Fig: Structure of bacteriophage (tadpole shaped)

- Head: icosahedral with DNA core. Neck: separates head and tail. Tail: hollow tube inside a contractile sheath; lysozyme at tip. Base plate: about 6 tail fibres and tail pins for attachment.

Q4. [SLO 5, 6, 7] Justify that a virus must have a host cell. How does it survive inside the host and outside it?

Ans:

- Host needed: obligate intracellular parasite; lacks enzymes and ribosomes; cannot multiply outside a living cell.
- Inside host: hides from immune system, changes genetic make-up so vaccines / antibodies fail, capsid protects genome from nucleases.
- Outside host: dormant, needs no food, no metabolism; genome stays viable for long; only pH and temperature matter; enveloped viruses die sooner than non-enveloped.

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Q5. [SLO 8] Describe the lytic and lysogenic life cycles of a bacteriophage.

Ans:

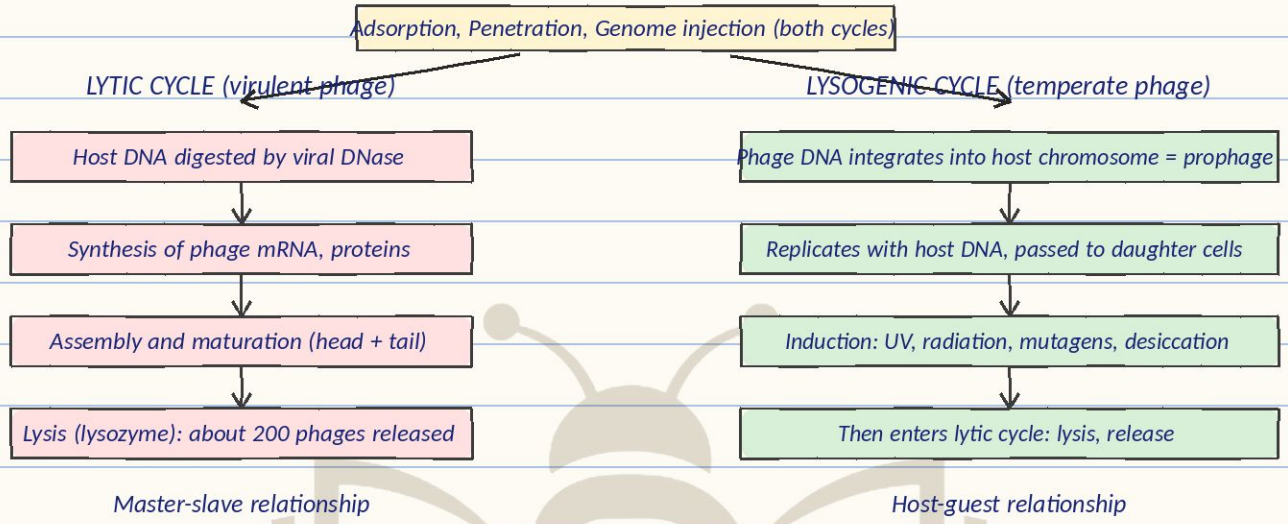


Fig: Lytic and lysogenic life cycles of bacteriophage

- *Common steps: adsorption, penetration, genome injection.*
- ***Lytic:** host DNA destroyed, phage components made, assembly and maturation, lysis; about 200 phages in 20-25 min.*
- ***Lysogenic:** DNA integrates as prophage, replicates with host; induction by UV, radiation, mutagens, desiccation triggers lytic cycle.*

Q6. [SLO 9] Outline the usage of bacteriophages in genetic engineering.

Ans:

- *Phage DNA as vector (besides plasmids) to make a genomic library.*
- *Phage therapy: engineered phages kill pathogenic bacteria while sparing normal flora (narrow spectrum), an advantage over antibiotics.*
- *Antibacterial nano medicine: phages as targeted drug-delivery vehicles.*

Q7. [SLO 4] Describe the structure of HIV with a labelled diagram.

Ans:

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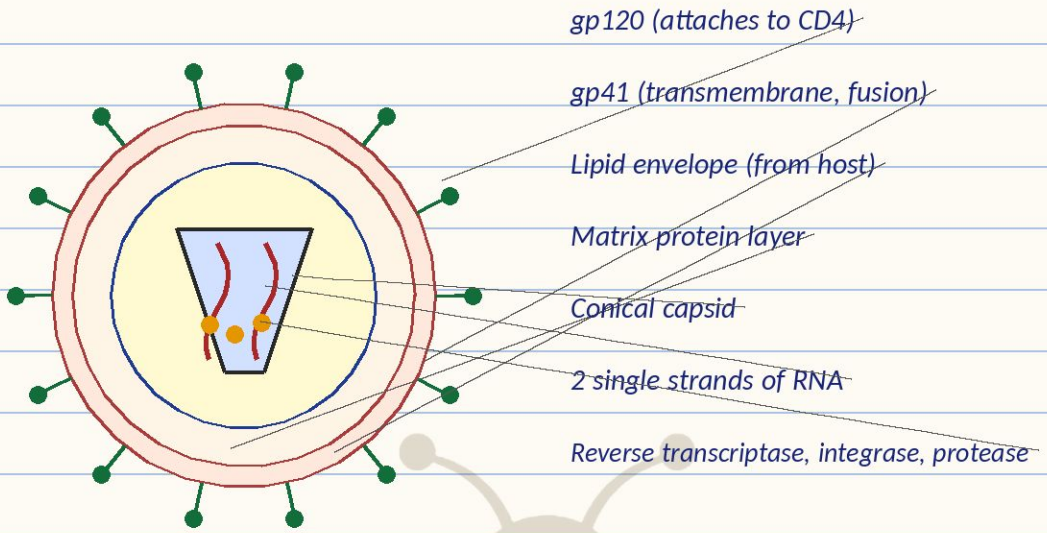


Fig: Structure of HIV (retrovirus)

- *Retrovirus, spherical; lipid envelope with gp120 and gp41 spikes; matrix protein layer; conical capsid; core with two ssRNA strands, reverse transcriptase, integrase and protease.*

Q8. [SLO 10] Explain the life cycle of HIV.

Ans:

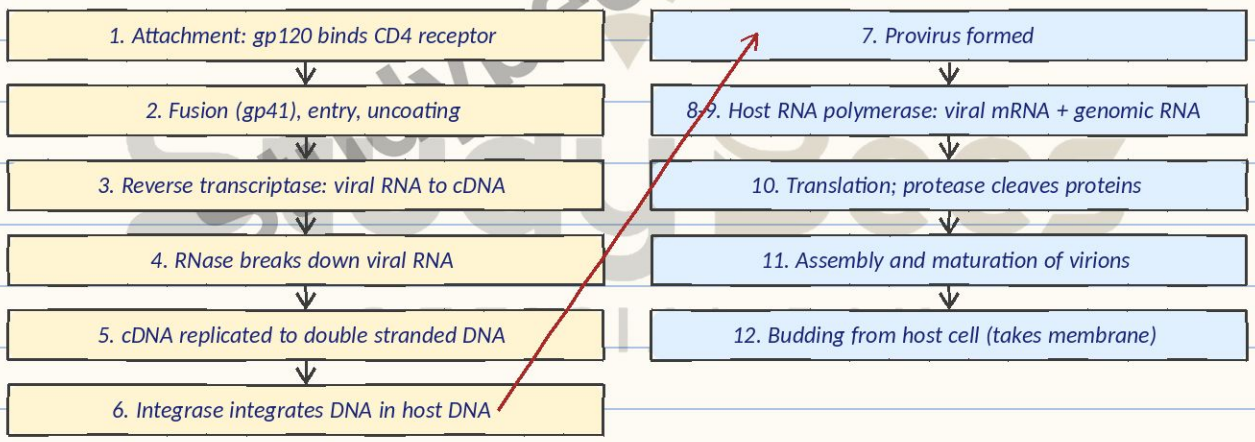


Fig: Life cycle of HIV

- *Attachment (gp120-CD4), fusion (gp41), uncoating, reverse transcription to cDNA, RNase destroys RNA, double stranded DNA, integration (integrase) as provirus, transcription by host RNA polymerase, translation, protease cleavage, assembly, budding.*

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Q9. [SLO 11, 12] Justify the name of HIV by establishing T-helper cells as basis of immune system. Reason out the specificity of HIV for host cells.

Ans:

- *Helper T cells activate B cells (antibodies) and cytotoxic T cells (kill infected and cancer cells) and enhance other immune cells.*
- *HIV reduces helper T cells: immune deficiency in humans = Human Immunodeficiency Virus.*
- *Specificity: gp120 binds only the CD4 receptor present on helper T cells, macrophages and brain cells.*

Q10. [SLO 13, 14] List the symptoms of AIDS and explain opportunistic diseases.

Ans:

Stage	***Symptoms***
Asymptomatic carrier	*Fever, chills, aches, swollen glands, rash; then 9 months or more without symptoms; highly infectious; antibody test positive*
ARC	*Persistent swollen glands, night sweats, cough, flu, diarrhoea, memory loss, depression*
Full blown AIDS	*Severe weight loss, weakness, opportunistic infections*

- *Opportunistic diseases: HIV only destroys T cells; the weak immune system allows skin cancer, fungal, viral, gastrointestinal, respiratory, nervous system and eye diseases.*

Q11. [SLO 15, 16] Describe the treatment of AIDS and the control measures against transmission of HIV.

Ans:

- *ART (antiretroviral therapy): combination of medicines; not a cure; controls the virus, prolongs life, reduces transmission; limits: toxicity, cost, strict daily pills.*
- *Control: avoid used syringes and needles; screen blood for transfusion; do not*

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share toothbrushes, blades, towels; careful at barbers; sterilize surgical instruments; avoid illicit sexual activity; infected mothers should not feed babies.

- *No transmission by handshakes, hugging, coughing, sneezing, swimming pools, toilets, door knobs, mosquitoes.*

Q12. [SLO 17, 18] Describe the causative agent, symptoms, treatment, prevention and transmission of hepatitis C and herpes.

Ans:

Hepatitis C	***Herpes***
Hepatitis C virus	*Herpes simplex virus (type 1 oral; type 2 genital)*
Only via blood	*Saliva, contact with sores (type 1); sexual contact (type 2)*
Symptoms like hepatitis B	*Painful fluid-filled blisters, tingling / itching first*
Alpha interferon + ribavirin	*Antiviral drugs*
No vaccine; screen blood	*Avoid contact with affected area*

Q13. [SLO 17, 18] Describe the causative agent, symptoms, treatment, prevention and transmission of polio and cotton leaf curl disease.

Ans:

- ***Polio:*** *polio virus (enterovirus); faecal-oral route; replicates in oropharynx and intestine, spreads to blood and CNS, kills motor neurons: permanent paralysis; no antiviral, physiotherapy; prevention by Salk (IPV) and Sabin (OPV) vaccines.*
- ***Cotton leaf curl:*** *Begomoviruses; whitefly vector; downward cupping of young leaves then cup-shaped leaf-like structures; control by insecticide against whitefly.*

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Q14. [SLO 19, 20] Describe the structure of prions and viroids and list the diseases they cause.

Ans:

Prions	Viroids
Infectious protein, no nucleic acid	Short circular ssRNA, no coat
Single protein of a cellular gene; resistant to UV, heat	No protein coded; RNA polymerase II replication; some ribozymes
Kuru, mad cow disease; no cure	Potato spindle tuber, cucumber pale fruit; hepatitis D (book)

Q15. [SLO 21, 22] Interpret how viral infections cause global economic loss. Describe the limitations of the vaccine for common cold / flu.

Ans:

- Economic loss: COVID-19 lockdowns, closed companies, job loss, hit on services, agriculture, tourism, airlines, trade and GDP; world cost over 8.5 trillion dollars; even weak flu pandemic (H1N1, 2009) could cut global GDP by about 300 billion dollars.
- Vaccine limits: over 200 viruses cause colds; flu strains constantly change; immune response changes over time, so earlier vaccines fail against new strains.

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